Microeconomic Industry Review of Enterprise Generative AI Software

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StraighterLine

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A growing market can still be a difficult place to earn a profit. This review evaluates enterprise generative artificial intelligence (AI) software, including business assistants, coding tools, and model application programming interfaces (APIs). It considers an application startup that purchases model access instead of training a frontier model. I assign the industry a B investment rating for 2027–2028. Demand and employment prospects are favorable, but computing expenses, competition, and uncertain cash flows prevent a confident A rating under the required low-cost and high-return criteria.

Supply and Demand

Enterprise buyers use AI to reduce the time spent writing, coding, and answering customer questions. Menlo Ventures (2025) estimates that enterprise generative AI spending rose from $11.5 billion in 2024 to $37 billion in 2025 (Figure 1). These are modeled estimates rather than audited industry revenues. Its U.S. survey and investment interests limit generalizability.

Productivity improvements raise willingness to pay and shift demand to the right. Noy and Zhang (2023) found that professionals using ChatGPT completed selected writing tasks 40% faster, with quality improving 18%. A buyer acting in rational self-interest compares those savings with subscription, training, and checking costs. Scarce budgets create an opportunity cost: spending on AI leaves less money for staff or other software. Additional AI seats may yield diminishing marginal utility once the most useful workflows are covered.

Supply expands as better models and cheaper inference make more applications commercially viable. The Stanford Institute for Human-Centered Artificial Intelligence (Stanford HAI, 2025) reports that the price of querying a model at roughly GPT-3.5 benchmark

performance fell from $20 to $0.07 per million tokens between November 2022 and October 2024 (Figure 2). This comparison holds capability approximately constant; it does not measure the cost of every AI task. Lower input costs shift application supply rightward. Demand and supply expansion increase equilibrium quantity, while their relative sizes determine the price change.

Reliable industry-wide price elasticity estimates are unavailable in the sources reviewed. Generic assistants probably face more elastic demand because buyers can switch among substitutes. Specialized products with costly integrations may face less elastic demand. As an illustrative assumption, elasticity of −1.5 means a 10% price increase would reduce quantity demanded approximately 15%, lowering revenue about 6.5%. This scenario is not an empirical estimate. It explains why raising prices can hurt profits even when the market is growing.

Factor Markets and Production Costs

Production combines skilled labor, computing capacity, software, and customer data to generate a usable service. Developers integrate models, test outputs, and maintain security; sales and support staff help customers adopt the product. The U.S. Bureau of Labor Statistics (BLS, 2026) reports a May 2025 median software developer wage of $135,980 and projects 10% growth in developer employment during 2025–2035. This occupational outlook supports the labor-growth case, but it is not an AI startup employment forecast.

Table 1 reports factor prices. Two developers at the national median would cost $271,960 annually before benefits. Renting computing capacity reduces equipment purchases, although it creates recurring expenses. An API-based startup can avoid reserving a dedicated GPU server.

Office rent, laptops, insurance, and data licensing still require a separate budget; their costs depend on location and operating choices.

Development salaries and initial integration are largely fixed over a short planning period. Token usage, storage, and transaction support vary with output. Anthropic's listed Sonnet 4.6 rates imply a $6 token bill for one million input tokens and 200,000 output tokens, excluding other services (Anthropic, n.d.). Thus, distributing software cheaply does not make additional AI usage free. As sales grow, fixed costs spread across customers and average fixed cost falls. Heavy usage or poor capacity planning can nevertheless raise marginal cost.

AI can also raise the marginal product of labor. Brynjolfsson et al. (2025) found a 15% average productivity increase among 5,172 customer-support agents. Such gains can expand a firm's production possibilities, but they do not establish equivalent savings in software development. With computing capacity fixed, adding workers eventually produces diminishing marginal returns. Specialization and better tools help; hiring alone cannot remove every bottleneck. Acemoglu (2025) cautions that large gains in selected tasks need not translate into equally large economy-wide gains.

Market Structure and Profitability

The market has two distinct layers. Foundation model APIs resemble a differentiated oligopoly, while application software has more entrants and features of monopolistic competition. OpenAI, Anthropic, and Google supply leading models. Application competitors include Microsoft Copilot, Cursor, Salesforce Agentforce, and specialist vendors. Their products differ in accuracy, integration, and intended customers. Neither layer fits perfect competition, and neither is a single-firm monopoly.

Figure 3 shows estimated API spending shares, not application subscription shares. The three leading model providers' combined share increased from 69% in 2023 to 88% in 2025. The remaining providers are grouped because individual shares are unavailable (Menlo Ventures, 2025). Reporting precise shares for every application vendor would require data these sources do not provide. Korinek and Vipra (2025) explain how scale economies in computing, data, and talent can reinforce concentration. Dependence on a few suppliers gives application startups exposure to contract and pricing changes.

Profitability evidence is less complete than spending evidence. Microsoft Cloud's gross margin declined from 71% in fiscal 2024 to 69% in fiscal 2025, with Microsoft identifying AI infrastructure expansion as a source of pressure (Microsoft, 2024, 2025). These figures cover a broader cloud business and cannot stand in for AI startup margins. Comparable audited standalone margins for private model providers are unavailable in the reviewed sources. Revenue growth alone therefore cannot establish industry profitability.

Market failures also affect returns. Buyers may struggle to assess accuracy before purchase, creating information asymmetry. Data misuse can impose costs on people outside a transaction. Independent evaluations, clear disclosures, consent controls, and human review can reduce these problems. Interoperability can limit lock-in, although compliance and verification add operating costs.

Two Year Forecast and Investment Recommendation

For 2027–2028, my base case is rising paid usage accompanied by stronger price competition. Table 2 illustrates this with a 2026 revenue index of 100, assumed annual usage growth of 30%, and annual price declines of 10%. Revenue reaches 117 in 2027 and 136.9 in

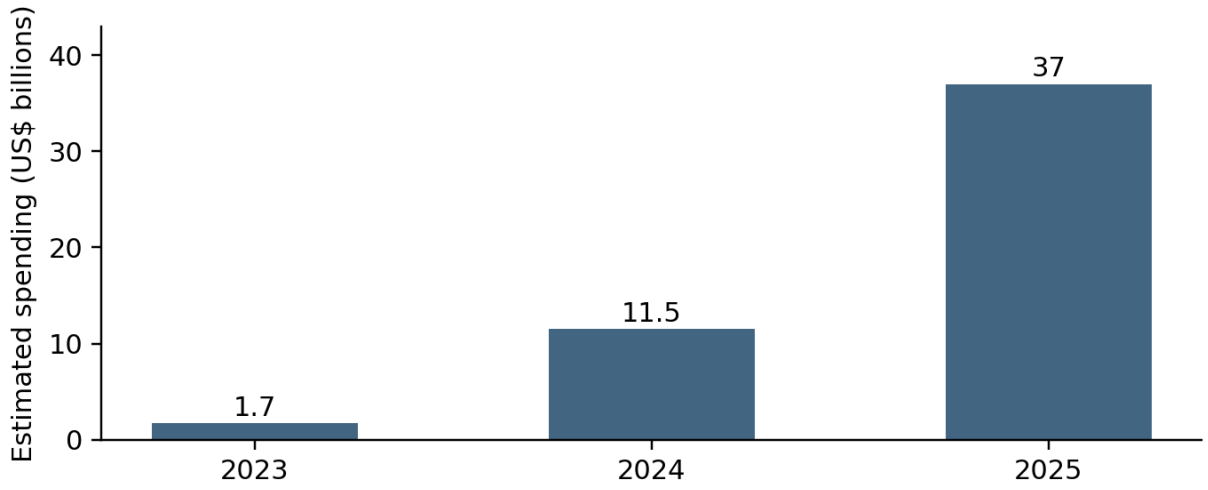
2028. These assumptions are planning scenarios, not published market forecasts. Slower adoption or faster price erosion would weaken the result.

Financing must be included before recommending a venture. An illustrative $200,000 interest-only loan at 10% costs $20,000 annually. At a 60% contribution margin, covering that interest and two median developer salaries requires about $486,600 in annual revenue, before other fixed expenses, taxes, or principal repayment. The loan rate and margin are assumptions, not observed industry averages. Positive cash flow would also need to justify the original investment and investors' required return.

I recommend selective investment in applications with paying customers, measurable savings, and controlled usage costs. The B rating recognizes strong demand while withholding the low-cost, high-return judgment needed for an A. A startup could earn an A after demonstrating customer retention, positive cash flow after financing costs, and resilience to supplier price changes. Until then, growth is a reason to investigate, not sufficient evidence to invest.

Figure 1

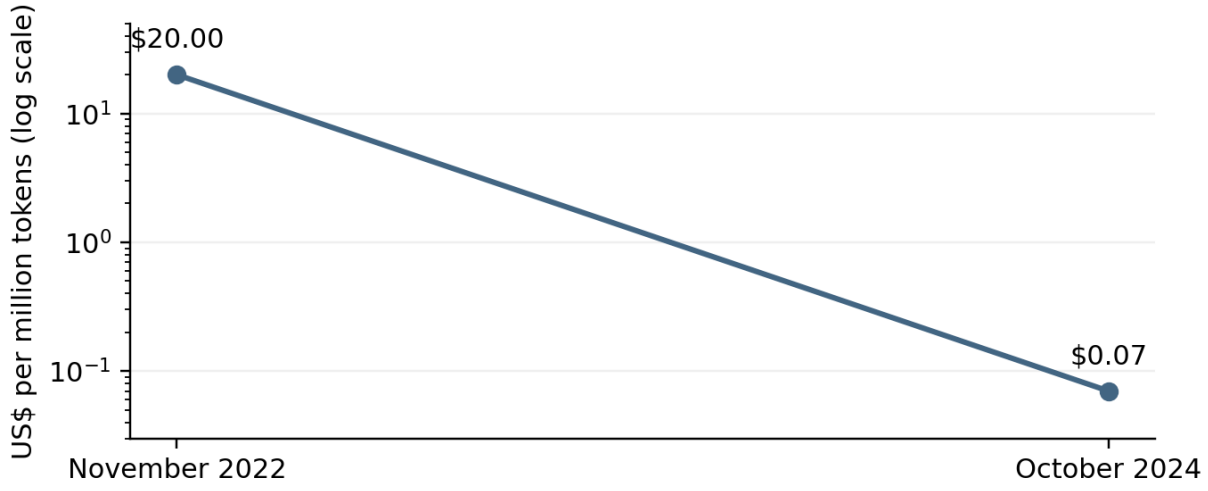
Estimated Enterprise Generative AI Spending



Note. Data from Menlo Ventures (2025). Revised estimates exclude inference and model serving.

Figure 2

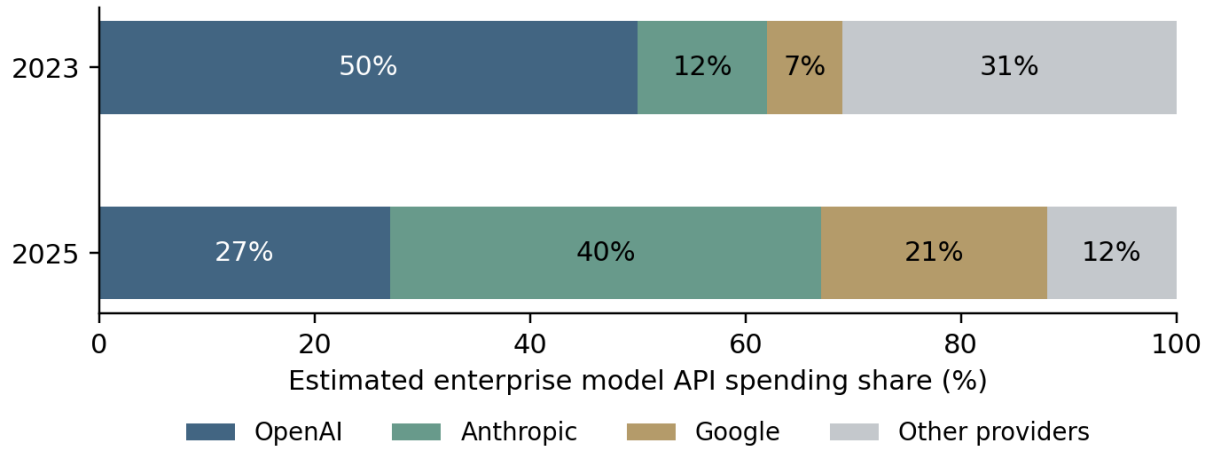
Query Prices at Approximately GPT 3.5 Benchmark Performance



Note. Data from Stanford HAI (2025). Logarithmic scale; matched capability does not imply identical models or workloads.

Figure 3

Enterprise Foundation Model API Market Shares



Note. Data from Menlo Ventures (2025). Weighted API usage estimates spending shares; these are not application market shares.

Table 1

Observed Labor and Computing Prices

Input	Price	Basis
Software developer	$135,980/year	U.S. median, May 2025
QA analyst or tester	$104,300/year	U.S. median, May 2025
AWS p4d.24xlarge	$13.57/hour	8 A100 GPUs; capacity block
Sonnet 4.6 API	$3 input; $15 output	Per million tokens

Note. Sources: BLS (2026), Amazon Web Services (n.d.), and Anthropic (n.d.). AWS rate is for Northern Virginia and excludes ancillary charges. Listed vendor prices accessed October 8, 2026.

Table 2

Illustrative Two Year Revenue Forecast

Measure	2026 base	2027	2028
Paid usage index	100.0	130.0	169.0
Realized price index	100.0	90.0	81.0
Revenue index	100.0	117.0	136.9
Demand and supply	Baseline	Both expand	Both expand

Note. Author's scenario. Annual usage growth is 30%; price falls 10%. Revenue equals usage × price / 100. No observed 2026 revenue or industry forecast is implied.

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